NTRK3 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes NTRK3 gene fusions from the msk_impact_50k_2026 study, covering 58 fusion events. 55/56 fusions with known frame status (98.2%) were in-frame. 55/57 fusions (96.5%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test (p=0.0555556, not statistically significant at alpha=0.05). No literature benchmark is configured for NTRK3.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 58 analyzed fusion events, identifying 4 distinct fusion partner genes. The most frequent partner was ETV6, observed in 55 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 54/55 (98.2%) of in-frame fusions retained the domain, p=0.0555556 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.001998.

Domain disruption

Domain-disruption analysis was skipped: NTRK3 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Junction concentration

60% of 35 fusions with this gene as the 3' partner join at exon 14, against 24% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=0.00019998, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 36 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 155 aa (exon 5/6 boundary; label-permutation p=0.021978). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is -5 aa from the nearest configured domain boundary at 160 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=35) and "not_retained" (n=1). For Frame_status, retained: 34/34 (100.0%, 95% CI 89.8%-100.0%); not_retained: 0/1

(0.0%, 95% CI 0.0%-79.3%). Welch's t-test comparing Tumor_variant_count between groups gave means of 57.4 vs 7, p=unavailable.

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for NTRK3 in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (33 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 2 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: NTRK3 has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: NTRK3 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (18 row(s) in window_scan, 4 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ETV6	55	0.9482758620689655	0.26994040818153375	None	0.0	0.5525173801119567	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.5090244983465859	1
DOT1L	1	0.017241379310344827	0.26994040818153375	None	1.0	0.5525173801119567	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.3473865673121032	2
VPS39	1	0.017241379310344827	0.26994040818153375	None	None	0.5525173801119567	confidence_certainty, domain_retention, recurrence	0.19678346746105008	3
ZNF710	1	0.017241379310344827	0.26994040818153375	None	0.0940573253761453	0.5525173801119567	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.17752231582013045	4

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
155	0	35	1	0	0.0	0.027777777777777776	1.5563025007672873
466	13	22	1	0	0.0	0.3888888888888889	0.41017446508904926
472	14	21	1	0	0.0	0.4166666666666667	0.38021124171160603

domain_retention tables

frame_domain_contingency_table

34	1
0	1

frame_domain_contingency_tables_by_tumor_type

stratum	table
BRCNOS	[1, 0], [0, 0]
COAD	[4, 0], [0, 0]

stratum	table
CUP	[1, 0], [0, 0]
FIBS	[1, 0], [0, 0]
GB	[0, 1], [0, 0]
HNMASC	[9, 0], [0, 0]
IDC	[1, 0], [0, 0]
IMT	[1, 0], [0, 0]
JSCB	[2, 0], [0, 0]
PAAD	[3, 0], [0, 1]
PANET	[1, 0], [0, 0]
PRNET	[1, 0], [0, 0]
SAAD	[1, 0], [0, 0]
SACA	[1, 0], [0, 0]
SARCNOS	[1, 0], [0, 0]
SDCA	[1, 0], [0, 0]
THPA	[4, 0], [0, 0]
THPD	[1, 0], [0, 0]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	36	35	0	1	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
14	35	37	0.9459459459459459

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
DOT1L	1	1	1	1
ETV6	55	55	55	34
VPS39	1	1	1	1

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
ZNF710	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
15	88505000	88506000	3	EVT-P-0002671-T01-IM3-34, EVT-P-0035767-T01-IM6-47, EVT-P-0055703-T01-IM6-13
15	88651000	88652000	2	EVT-P-0026390-T01-IM6-20, EVT-P-0035552-T01-IM6-46
15	88488000	88489000	1	EVT-P-0016507-T01-IM6-24
15	88492000	88493000	1	EVT-P-0058408-T01-IM6-15
15	88497000	88498000	1	EVT-P-0040355-T01-IM6-12
15	88498000	88499000	1	EVT-P-0006480-T01-IM5-45
15	88501000	88502000	1	EVT-P-0013196-T02-IM6-56
15	88506000	88507000	1	EVT-P-0002978-T02-IM5-39
15	88512000	88513000	1	EVT-P-0027006-T01-IM6-36
15	88513000	88514000	1	EVT-P-0010668-T01-IM5-55
15	88514000	88515000	1	EVT-P-0018060-T01-IM6-28
15	88516000	88517000	1	EVT-P-0033921-T02-IM6-6
15	88522000	88523000	1	EVT-P-0032148-T04-IM6-5
15	88544000	88545000	1	EVT-P-0025507-T01-IM6-31
15	88548000	88549000	1	EVT-P-0014872-T02-IM6-21
15	88550000	88551000	1	EVT-P-0021974-T01-IM6-29
15	88554000	88555000	1	EVT-P-0061265-T01-IM7-16
15	88556000	88557000	1	EVT-P-0042503-T01-IM6-11
15	88558000	88559000	1	EVT-P-0013664-T02-IM5-2
15	88564000	88565000	1	EVT-P-0064964-T01-IM7-1
15	88576000	88577000	1	EVT-P-0010556-T01-IM5-70
15	88581000	88582000	1	EVT-P-0010562-T01-IM5-53
15	88588000	88589000	1	EVT-P-0010264-T01-IM5-52
15	88592000	88593000	1	EVT-P-0007911-T02-IM5-51
15	88594000	88595000	1	EVT-P-0037610-T01-IM6-10
15	88600000	88601000	1	EVT-P-0007772-T01-IM5-48
15	88611000	88612000	1	EVT-P-0026286-T01-IM6-33

chromosome	bin_start	bin_end	n_events	event_ids
15	88613000	88614000	1	EVT-P-0025987-T01-IM6-32
15	88624000	88625000	1	EVT-P-0056156-T02-IM7-14
15	88632000	88633000	1	EVT-P-0004737-T01-IM5-44
15	88657000	88658000	1	EVT-P-0026696-T01-IM6-35
15	88662000	88663000	1	EVT-P-0023257-T01-IM6-30
15	88683000	88684000	1	EVT-P-0058230-T02-IM7-19

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
529	21	21	75862	False
466	13	13	80532	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
three_prime	14	21	35	0.6	0.24295121694236646	5.134613335190334
three_prime	13	14	35	0.4	0.24605942471815404	1.4947930796250748
five_prime	5	1	1	1.0	0.025924560109577496	1.5862886040268265

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
266	466	200	13	23	13	22	None	1.0	-0.0	EVT-P-0004737-T01-IM5-44, EVT-P-0007772-T01-IM5-48, EVT-P-0007911-T02-IM5-51, EVT-P-0010264-T01-IM5-52, EVT-P-0010562-T01-IM5-53, ... (13 total)
272	472	200	14	22	14	21	None	1.0	-0.0	EVT-P-0004737-T01-IM5-44, EVT-P-0007772-T01-IM5-48, EVT-P-0007911-T02-IM5-51, EVT-P-0010264-T01-IM5-52, EVT-P-0010556-T01-IM5-70, ... (14 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
366	466	100	13	23	13	22	None	1.0	-0.0	EVT-P-0004737-T01-IM5-44, EVT-P-0007772-T01-IM5-48, EVT-P-0007911-T02-IM5-51, EVT-P-0010264-T01-IM5-52, EVT-P-0010562-T01-IM5-53, ... (13 total)
372	472	100	14	22	14	21	None	1.0	-0.0	EVT-P-0004737-T01-IM5-44, EVT-P-0007772-T01-IM5-48, EVT-P-0007911-T02-IM5-51, EVT-P-0010264-T01-IM5-52, EVT-P-0010556-T01-IM5-70, ... (14 total)
416	466	50	13	23	13	22	None	1.0	-0.0	EVT-P-0004737-T01-IM5-44, EVT-P-0007772-T01-IM5-48, EVT-P-0007911-T02-IM5-51, EVT-P-0010264-T01-IM5-52, EVT-P-0010562-T01-IM5-53, ... (13 total)
422	472	50	14	22	14	21	None	1.0	-0.0	EVT-P-0004737-T01-IM5-44, EVT-P-0007772-T01-IM5-48, EVT-P-0007911-T02-IM5-51, EVT-P-0010264-T01-IM5-52, EVT-P-0010556-T01-IM5-70, ... (14 total)
441	466	25	13	23	13	22	None	1.0	-0.0	EVT-P-0004737-T01-IM5-44, EVT-P-0007772-T01-IM5-48, EVT-P-0007911-T02-IM5-51, EVT-P-0010264-T01-IM5-52, EVT-P-0010562-T01-IM5-53, ... (13 total)
447	472	25	14	22	14	21	None	1.0	-0.0	EVT-P-0004737-T01-IM5-44, EVT-P-0007772-T01-IM5-48, EVT-P-0007911-T02-IM5-51, EVT-P-0010264-T01-IM5-52, EVT-P-0010556-T01-IM5-70, ... (14 total)
466	491	25	14	22	14	21	None	1.0	-0.0	EVT-P-0004737-T01-IM5-44, EVT-P-0007772-T01-IM5-48, EVT-P-0007911-T02-IM5-51, EVT-P-0010264-T01-IM5-52, EVT-P-0010556-T01-IM5-70, ... (14 total)
466	516	50	14	22	14	21	None	1.0	-0.0	EVT-P-0004737-T01-IM5-44, EVT-P-0007772-T01-IM5-48, EVT-P-0007911-T02-IM5-51, EVT-P-0010264-T01-IM5-52, EVT-P-0010556-T01-IM5-70, ... (14 total)
472	572	100	22	14	22	13	None	0.3888888888888889	0.41017446508904926	EVT-P-0002671-T01-IM3-34, EVT-P-0002978-T02-IM5-39, EVT-P-0006480-T01-IM5-45, EVT-P-0010556-T01-IM5-70, EVT-P-0010668-T01-IM5-55, ... (22 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
472	672	200	22	14	22	13	None	0.3888888888888889	0.41017446508904926	EVT-P-0002671-T01-IM3-34, EVT-P-0002978-T02-IM5-39, EVT-P-0006480-T01-IM5-45, EVT-P-0010556-T01-IM5-70, EVT-P-0010668-T01-IM5-55, ... (22 total)
479	529	50	21	15	21	14	None	0.4166666666666667	0.38021124171160603	EVT-P-0002671-T01-IM3-34, EVT-P-0002978-T02-IM5-39, EVT-P-0006480-T01-IM5-45, EVT-P-0010668-T01-IM5-55, EVT-P-0013196-T02-IM6-56, ... (21 total)
504	529	25	21	15	21	14	None	0.4166666666666667	0.38021124171160603	EVT-P-0002671-T01-IM3-34, EVT-P-0002978-T02-IM5-39, EVT-P-0006480-T01-IM5-45, EVT-P-0010668-T01-IM5-55, EVT-P-0013196-T02-IM6-56, ... (21 total)
529	554	25	21	15	21	14	None	0.4166666666666667	0.38021124171160603	EVT-P-0002671-T01-IM3-34, EVT-P-0002978-T02-IM5-39, EVT-P-0006480-T01-IM5-45, EVT-P-0010668-T01-IM5-55, EVT-P-0013196-T02-IM6-56, ... (21 total)
529	579	50	21	15	21	14	None	0.4166666666666667	0.38021124171160603	EVT-P-0002671-T01-IM3-34, EVT-P-0002978-T02-IM5-39, EVT-P-0006480-T01-IM5-45, EVT-P-0010668-T01-IM5-55, EVT-P-0013196-T02-IM6-56, ... (21 total)
529	629	100	21	15	21	14	None	0.4166666666666667	0.38021124171160603	EVT-P-0002671-T01-IM3-34, EVT-P-0002978-T02-IM5-39, EVT-P-0006480-T01-IM5-45, EVT-P-0010668-T01-IM5-55, EVT-P-0013196-T02-IM6-56, ... (21 total)
529	729	200	21	15	21	14	None	0.4166666666666667	0.38021124171160603	EVT-P-0002671-T01-IM3-34, EVT-P-0002978-T02-IM5-39, EVT-P-0006480-T01-IM5-45, EVT-P-0010668-T01-IM5-55, EVT-P-0013196-T02-IM6-56, ... (21 total)

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
472	572	100	22	14	22	13	None	0.3888888888888889	0.41017446508904926	EVT-P-0002671-T01-IM3-34, EVT-P-0002978-T02-IM5-39, EVT-P-0006480-T01-IM5-45, EVT-P-0010556-T01-IM5-70, EVT-P-0010668-T01-IM5-55, ... (22 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
504	529	25	21	15	21	14	None	0.4166666666666667	0.38021124171160603	EVT-P-0002671-T01-IM3-34, EVT-P-0002978-T02-IM5-39, EVT-P-0006480-T01-IM5-45, EVT-P-0010668-T01-IM5-55, EVT-P-0013196-T02-IM6-56, ... (21 total)
441	466	25	13	23	13	22	None	1.0	-0.0	EVT-P-0004737-T01-IM5-44, EVT-P-0007772-T01-IM5-48, EVT-P-0007911-T02-IM5-51, EVT-P-0010264-T01-IM5-52, EVT-P-0010562-T01-IM5-53, ... (13 total)
447	472	25	14	22	14	21	None	1.0	-0.0	EVT-P-0004737-T01-IM5-44, EVT-P-0007772-T01-IM5-48, EVT-P-0007911-T02-IM5-51, EVT-P-0010264-T01-IM5-52, EVT-P-0010556-T01-IM5-70, ... (14 total)

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0064964-T01-IM7-1	P-0064964-T01-IM7	P-0064964	Breast Cancer	JSCB	ETV6-NTRK3	ETV6	in-frame	three_prime	88564682	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12032310::chr15:g.88564682) | Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0013664-T02-IM5-2	P-0013664-T02-IM5	P-0013664	Soft Tissue Sarcoma	FIBS	ETV6-NTRK3	ETV6	in-frame	three_prime	88558191	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) Fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q26.1)(chr12:g.12032077::chr15:g.88558191) | Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013664-T04-IM6-3	P-0013664-T04-IM6	P-0013664	Soft Tissue Sarcoma	FIBS	ETV6-NTRK3	ETV6	in-frame	three_prime	88558191	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032077::chr15:g.88558191) | Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0033921-T05-IM7-4	P-0033921-T05-IM7	P-0033921	Breast Cancer	JSCB	ETV6-NTRK3	ETV6	in-frame	three_prime	88516534	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12031161::chr15:g.88516534) | Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032148-T04-IM6-5	P-0032148-T04-IM6	P-0032148	Salivary Gland Cancer	HNMASC	ETV6-NTRK3	ETV6	in-frame	three_prime	88522437	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12037217::chr15:g.88522437) | Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0033921-T02-IM6-6	P-0033921-T02-IM6	P-0033921	Breast Cancer	JSCB	ETV6-NTRK3	ETV6	in-frame	three_prime	88516534	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12031161::chr15:g.88516534) | Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0033921-T04-IM6-7	P-0033921-T04-IM6	P-0033921	Breast Cancer	JSCB	ETV6-NTRK3	ETV6	in-frame	three_prime	88516534	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12031161::chr15:g.88516534) | Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0035767-T06-IM7-8	P-0035767-T06-IM7	P-0035767	Pancreatic Cancer	PAAC	ETV6-NTRK3	ETV6	in-frame	three_prime	88505787	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12029143::chr15:g.88505787) | Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0035767-T04-IM6-9	P-0035767-T04-IM6	P-0035767	Pancreatic Cancer	PAAD	ETV6-NTRK3	ETV6	in-frame	three_prime	88505787	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12029143::chr15:g.88505787) | Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0037610-T01-IM6-10	P-0037610-T01-IM6	P-0037610	Thyroid Cancer	THPA	ETV6-NTRK3	ETV6	in-frame	three_prime	88594532	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) rearrangement: t(12;15)(p13.2;q25.3)(chr12:g.12014663::chr15:g.88594532) | Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0042503-T01-IM6-11	P-0042503-T01-IM6	P-0042503	Colorectal Cancer	COAD	ETV6-NTRK3	ETV6	in-frame	three_prime	88556067	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12032802::chr15:g.88556067) | Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0040355-T01-IM6-12	P-0040355-T01-IM6	P-0040355	Colorectal Cancer	COAD	ETV6-NTRK3	ETV6	in-frame	three_prime	88497191	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12036111::chr15:g.88497191) | Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0055703-T01-IM6-13	P-0055703-T01-IM6	P-0055703	Cancer of Unknown Primary	CUP	ETV6-NTRK3	ETV6	in-frame	three_prime	88505980	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12024785::chr15:g.88505980) | Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0056156-T02-IM7-14	P-0056156-T02-IM7	P-0056156	Thyroid Cancer	THPA	ETV6-NTRK3	ETV6	in-frame	three_prime	88624274	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12011957::chr15:g.88624274) | Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0058408-T01-IM6-15	P-0058408-T01-IM6	P-0058408	Colorectal Cancer	COAD	ETV6-NTRK3	ETV6	in-frame	three_prime	88492156	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12035760::chr15:g.88492156) | Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0061265-T01-IM7-16	P-0061265-T01-IM7	P-0061265	Salivary Gland Cancer	HNMASC	ETV6-NTRK3	ETV6	in-frame	three_prime	88554122	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12033175::chr15:g.88554122) | Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0058230-T02-IM7-19	P-0058230-T02-IM7	P-0058230	Pancreatic Cancer	PAAD	NTRK3-DOT1L	DOT1L	out-of-frame	five_prime	88683244	5	155	True	154	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Protein kinase domain		NTRK3 (NM_001012338) - DOT1L (NM_032482) rearrangement: t(15;19)(q25.3;p13.3)(chr15:g.88683244::chr19:g.2227604) | Protein Fusion: out of frame {NTRK3:DOT1L}	NA
EVT-P-0026390-T01-IM6-20	P-0026390-T01-IM6	P-0026390	Thyroid Cancer	THPA	NTRK3-ETV6	ETV6	in-frame	three_prime	88651131	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12016863::chr15:g.88651131) | Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014872-T02-IM6-21	P-0014872-T02-IM6	P-0014872	Soft Tissue Sarcoma	IMT	NTRK3-ETV6	ETV6	in-frame	three_prime	88548029	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028946::chr15:g.88548029) | Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0014872-T03-IM6-22	P-0014872-T03-IM6	P-0014872	Soft Tissue Sarcoma	IMT	NTRK3-ETV6	ETV6	in-frame	three_prime	88548029	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028946::chr15:g.88548029) | Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002978-T03-IM5-23	P-0002978-T03-IM5	P-0002978	Salivary Gland Cancer	HNMASC	NTRK3-ETV6	ETV6	in-frame	three_prime	88506571	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1 to 5 fused to NTRK3 exons 15 to 20) : t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) | Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0016507-T01-IM6-24	P-0016507-T01-IM6	P-0016507	Salivary Gland Cancer	HNMASC	NTRK3-ETV6	ETV6	in-frame	three_prime	88488820	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12023709::chr15:g.88488820) | Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016507-T02-IM6-25	P-0016507-T02-IM6	P-0016507	Salivary Gland Cancer	HNMASC	NTRK3-ETV6	ETV6	in-frame	three_prime	88488820	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-14 fused with NTRK3 exons 6-20 including the kinase domain): t(12,15)(p13.2,q25.3)(chr12:g.12023709::chr15:g.88488820) | Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016507-T03-IM6-26	P-0016507-T03-IM6	P-0016507	Salivary Gland Cancer	HNMASC	NTRK3-ETV6	ETV6	in-frame	three_prime	88488820	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20: t(12;15)(p13.2;q25.3)(chr12:g.12023709::chr15:g.88488820) | Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0016507-T04-IM6-27	P-0016507-T04-IM6	P-0016507	Salivary Gland Cancer	HNMASC	NTRK3-ETV6	ETV6	in-frame	three_prime	88488820	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12023709::chr15:g.88488820) | Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0018060-T01-IM6-28	P-0018060-T01-IM6	P-0018060	Salivary Gland Cancer	HNMASC	NTRK3-ETV6	ETV6	in-frame	three_prime	88514531	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12037037::chr15:g.88514531) | Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0021974-T01-IM6-29	P-0021974-T01-IM6	P-0021974	Salivary Gland Cancer	SAAD	NTRK3-ETV6	ETV6	in-frame	three_prime	88550500	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028417::chr15:g.88550500) | Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0023257-T01-IM6-30	P-0023257-T01-IM6	P-0023257	Salivary Gland Cancer	HNMASC	NTRK3-ETV6	ETV6	in-frame	three_prime	88662190	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1 to 4 fused to NTRK3 exons 14 to 20) : t(12;15)(p13.2;q25.3)(chr12:g.12014998::chr15:g.88662190) | Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0025507-T01-IM6-31	P-0025507-T01-IM6	P-0025507	Breast Cancer	IDC	NTRK3-ETV6	ETV6	in-frame	three_prime	88544573	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12024078::chr15:g.88544573) | Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025987-T01-IM6-32	P-0025987-T01-IM6	P-0025987	Miscellaneous Brain Tumor	PRNET	NTRK3-ETV6	ETV6	in-frame	three_prime	88613114	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12028942::chr15:g.88613114) | Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0026286-T01-IM6-33	P-0026286-T01-IM6	P-0026286	Salivary Gland Cancer	SACA	NTRK3-ETV6	ETV6	in-frame	three_prime	88611476	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12030395::chr15:g.88611476) | Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002671-T01-IM3-34	P-0002671-T01-IM3	P-0002671	Colorectal Cancer	COAD	NTRK3-ETV6	ETV6	in-frame	three_prime	88505784	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			NA | Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0026696-T01-IM6-35	P-002669 6-T01-IM6	P-002669 6	Thyroid Cancer	THPA	NTRK3-ETV6	ETV6	in-frame	three_prime	88657925	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20):t(12;15)(p13.2;q25.3)(chr12:g.12007967::chr15:g.88657925) | Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027006-T01-IM6-36	P-0027006-T01-IM6	P-0027006	Soft Tissue Sarcoma	SARCNOS	NTRK3-ETV6	ETV6	in-frame	three_prime	88512343	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12032352::chr15:g.88512343) | Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0027006-T03-IM6-37	P-0027006-T03-IM6	P-0027006	Soft Tissue Sarcoma	FIBS	NTRK3-ETV6	ETV6	in-frame	three_prime	88512343	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12037037::chr15:g.88514531) | Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0027006-T04-IM6-38	P-0027006-T04-IM6	P-0027006	Soft Tissue Sarcoma	FIBS	NTRK3-ETV6	ETV6	in-frame	three_prime	88512343	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12032352::chr15:g.88512343) | Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0002978-T02-IM5-39	P-0002978-T02-IM5	P-0002978	Salivary Gland Cancer	HNMASC	NTRK3-ETV6	ETV6	in-frame	three_prime	88506571	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			NA | Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002978-T05-IM6-40	P-0002978-T05-IM6	P-0002978	Salivary Gland Cancer	HNMASC	NTRK3-ETV6	ETV6	in-frame	three_prime	88506571	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) | Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0002978-T06-IM6-41	P-0002978-T06-IM6	P-0002978	Salivary Gland Cancer	HNMASC	NTRK3-ETV6	ETV6	in-frame	three_prime	88506571	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) | Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013196-T04-IM6-42	P-0013196-T04-IM6	P-0013196	Breast Cancer	BRCNOS	NTRK3-ETV6	ETV6	in-frame	three_prime	88501357	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3)(chr12:g.12036072::chr15:g.88501357) | Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0033921-T03-IM6-43	P-0033921-T03-IM6	P-0033921	Breast Cancer	BRCNOS	NTRK3-ETV6	ETV6	in-frame	three_prime	88516534	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12031161::chr15:g.88516534) | Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004737-T01-IM5-44	P-0004737-T01-IM5	P-0004737	Pancreatic Cancer	PAAD	NTRK3-ETV6	ETV6	in-frame	three_prime	88632786	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) Fusion (ETV6 exons 1 to 4 fused to NTRK3 exons 13 to 20) : t(12;15)(p13.2;q25.3)(chr12:g.12006809::chr15:g.88632786) | Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0006480-T01-IM5-45	P-0006480-T01-IM5	P-0006480	Salivary Gland Cancer	HNMASC	NTRK3-ETV6	ETV6	in-frame	three_prime	88498145	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12029548::chr15:g.88498145) | Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0035552-T01-IM6-46	P-0035552-T01-IM6	P-0035552	Pancreatic Cancer	PAAD	NTRK3-ETV6	ETV6	in-frame	three_prime	88651251	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20): t(12;15)(p13.2;q25.3)(chr12:g.12020963::chr15:g.88651251) | Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0035767-T01-IM6-47	P-0035767-T01-IM6	P-0035767	Pancreatic Cancer	PANET	NTRK3-ETV6	ETV6	in-frame	three_prime	88505787	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion: t(12;15)(p13.2;q25.3)(chr12:g.12029143::chr15:g.88505787) | Protein Fusion: in frame {ETV6:NTRK3}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007772-T01-IM5-48	P-0007772-T01-IM5	P-0007772	Salivary Gland Cancer	HNMASC	NTRK3-ETV6	ETV6	in-frame	three_prime	88600844	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257::chr15:g.88600844) | Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0007772-T02-IM5-49	P-0007772-T02-IM5	P-0007772	Salivary Gland Cancer	HNMASC	NTRK3-ETV6	ETV6	in-frame	five_prime	88600844	12	466	True	465	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		Protein kinase domain		ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257::chr15:g.88600844) | Protein fusion: in frame (NTRK3-ETV6)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007772-T03-IM5-50	P-0007772-T03-IM5	P-0007772	Salivary Gland Cancer	HNMASC	NTRK3-ETV6	ETV6	in-frame	three_prime	88600844	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-4 fused with NTRK3 exons 14-20) : t(12;15)(p13.2;q25.3)(chr12:g.12020257::chr15:g.88600844) | Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0007911-T02-IM5-51	P-0007911-T02-IM5	P-0007911	Thyroid Cancer	THPD	NTRK3-ETV6	ETV6	in-frame	three_prime	88592430	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1 to 4 fused with NTRK3 exons 13-20): t(12;15)(p13.2;q25.3)(chr12:g.12007142::chr15:g.88592430) | Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0010264-T01-IM5-52	P-0010264-T01-IM5	P-0010264	Pancreatic Cancer	PAAD	NTRK3-ETV6	ETV6	in-frame	three_prime	88588255	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-4 fused in frame with NTRK3 exons 13-20): t(12;15)(p13.2;q25.3)(chr12:g.12014962::chr15:g.88588255) | Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0010562-T01-IM5-53	P-0010562-T01-IM5	P-0010562	Salivary Gland Cancer	SDCA	NTRK3-ETV6	ETV6	in-frame	three_prime	88581658	13	466	True	466	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 14-20): t(12;15)(p13.2; q25.1)(chr12:g.12028903::chr15:g.88581658) | Protein fusion: in frame (ETV6-NTRK3)	NA

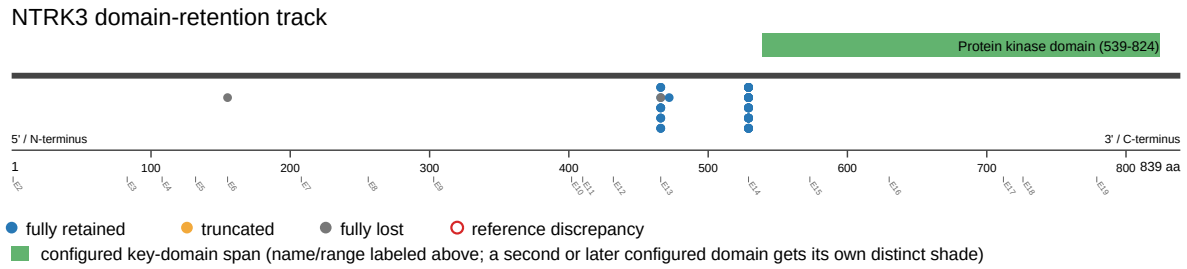
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002978-T04-IM5-54	P-0002978-T04-IM5	P-0002978	Salivary Gland Cancer	HNMASC	NTRK3-ETV6	ETV6	in-frame	three_prime	88506571	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) reciprocal fusion (ETV6 exons 1-5 fused with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) | Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0010668-T01-IM5-55	P-0010668-T01-IM5	P-0010668	Salivary Gland Cancer	HNMASC	NTRK3-ETV6	ETV6	in-frame	three_prime	88513282	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 with NTRK3 exons 15-20) : t(12;15)(p13.2;q25.3)(chr12:g.12029109::chr15:g.88513282) | Protein fusion: in frame (ETV6-NTRK3)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013196-T02-IM6-56	P-0013196-T02-IM6	P-0013196	Breast Cancer	BRCNOS	NTRK3-ETV6	ETV6	in-frame	three_prime	88501357	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3)(chr12:g.12036072::chr15:g.88501357) | Protein fusion: in frame (ETV6-NTRK3)	NA
EVT-P-0013196-T03-IM6-57	P-0013196-T03-IM6	P-0013196	Breast Cancer	BNNOS	NTRK3-ETV6	ETV6	in-frame	three_prime	88501357	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exon5 fused to NTRK3 exon15): t(12;15)(p13.2;q25.3)(chr12:g.12036072::chr15:g.88501357) | Protein fusion: in frame (ETV6-NTRK3)	NA

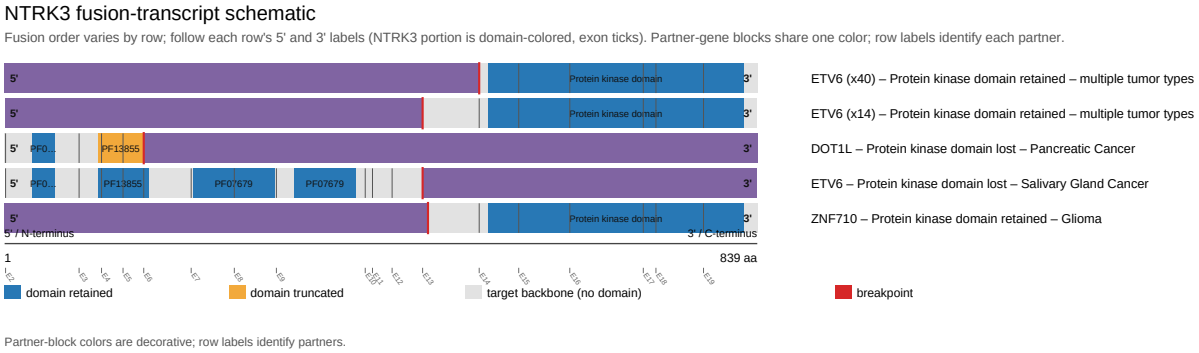
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002978-T07-IM6-58	P-0002978-T07-IM6	P-0002978	Salivary Gland Cancer	HNMASC	NTRK3-ETV6	ETV6	in-frame	three_prime	88506571	14	529	True	529	True	exact match	retained	1.0	False	Protein kinase domain			ETV6 (NM_001987) - NTRK3 (NM_001012338) fusion (ETV6 exons 1-5 fused in-frame to NTRK3 exons 15-20): t(12;15)(p13.2;q25.3)(chr12:g.12028576::chr15:g.88506571) | Protein Fusion: in frame {ETV6:NTRK3}	NA
EVT-P-0010556-T01-IM5-70	P-0010556-T01-IM5	P-0010556	Glioma	GB	NTRK3-ZNF710	ZNF710	unknown	three_prime	88576258	13	472	False	472	True	exact match	retained	1.0	False	Protein kinase domain			NTRK3 (NM_001012338) - ZNF710 (NM_198526) rearrangement: c.-28-10227:ZNF710_c.1415:NTRK3inv | Protein fusion: mid-exon (ZNF710-NTRK3)	NA

Figures

domain retention outliers



fusion schematic



intragenic deletion schematic

